

AMPL Training Series · Part 1 of 2



AMPL

The Modern AMPL Stack

From the legacy IDE to Python-first workflows for teaching, research, and the road to industry.

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THE HOUR AHEAD

Today's Agenda

- 01 **Why the stack moved**

- 02 **The AMPL Python API**

- 03 **Data & solver integration**

- 04 **Toolkit + live demo**

- 05 **Academic licensing**

Quick check — where are you?

AUDIENCE POLL · ZOOM

- A** Legacy IDE only
- B** Exploring Python tooling
- C** Already teaching with notebooks
- D** Mixed across courses

Drop a reaction or use the Zoom poll.

SECTION 01

Why the stack moved

SECTION 01 · WHY

What changed around AMPL

Python everywhere

The lingua franca of computational coursework — across OR, data science, and engineering.

Jupyter in the classroom

Narrative, code, and results in one artifact — the default form factor for assignments.

Cloud-first students

Chromebooks, no admin rights, the expectation that things just run in any browser tab.

Your AMPL hasn't changed. The world around it has.

SECTION 01 · WHY

Your .mod files still work

LEGACY AMPL · CLASSIC IDE

```
● ● ● PRODUCTION.MOD  
  
var x {P} >= 0;  
  
maximize Profit:  
    sum {p in P} margin[p] * x[p];  
  
subj to demand {p in P}:  
    x[p] <= cap[p];
```

MODERN STACK · JUPYTER CELL

```
● ● ● IN [3]: NOTEBOOK.IPYNB  
  
%%ampl_eval  
var x {P} >= 0;  
  
maximize Profit:  
    sum {p in P} margin[p] * x[p];  
  
subj to demand {p in P}:  
    x[p] <= cap[p];
```

The AMPL language is identical. What's new is everything wrapping it — Python, notebooks, package management, cloud.

A better toolbox

Editors

VS Code · amplpy · Jupyter

Use as much or as little of the modern stack as fits your course.

Cloud

Google Colab · MO-Book

Zero install. Runs in any browser tab.

Distribution

pip · Portal · amplbot

One command anywhere — Mac, Windows, Linux, Colab.

SECTION 02

What the modern toolbox looks like

SECTION 02 • THE PYTHON API

What amplpy does

AMPL ↔ PYTHON • LOAD DATA

```
LOAD_DATA.PY

from amplpy import AMPL
import pandas as pd

ampl = AMPL()
ampl.read('production.mod')

df = pd.read_csv('demand.csv')
ampl.set_data(df, 'P')
ampl.solve(solver='highs')
```

AMPL ↔ PYTHON • READ RESULTS

```
READ_RESULTS.PY

x = ampl.get_variable('x')
results = x.get_values().to_pandas()

results.plot.bar()
plt.title('Production plan')

print(ampl.get_objective('Profit').value())
```

Pandas in. Pandas out. The model is the only thing in AMPL syntax.

SECTION 02 · PYTHON API

Why this matters for teaching and research

Version control

Notebooks and .mod files in Git — full diff history on student assignments.

Parameter sweeps

Solve hundreds of variants from a Python loop. No hand-edited .dat files.

Sharable artifacts

One .ipynb runs on a laptop, a Google Colab tab, or a cloud instance.

Stop writing **.dat** files

⚠ CLASSICAL · PLANTS.DAT

```
● ● ●                                ⚠ PLANTS.DAT

set Plants :=
    Detroit  Pittsburgh  Atlanta;

param capacity :=
    Detroit      900
    Pittsburgh   1200
    Atlanta      800;
```

MODERN · PANDAS DATAFRAME

```
● ● ●                                IN [4]: NOTEBOOK.IPYNB

import pandas as pd
# change to amplpy
df = pd.read_csv("plants.csv")
ampl.set_data(df, "Plants")

# now drives the same model
```

Reach for pandas when your data lives in CSVs, databases, or APIs — not in **.dat** files.

SECTION 03

How AMPL plugs into Python

Installing solvers is just **pip** now

```
~/COURSE — TERMINAL  
  
$ python -m pip install amplpy --upgrade  
$ python -m amplpy.modules install highs gurobi xpress cplex  
$ python -m amplpy.modules activate <your-license-uuid>
```

No PATH wrangling

No separate downloads

Same command on Mac, Windows, Linux, Colab

Three commands. Same on Mac, Windows, Linux, and Colab.

Swap solvers without changing the model

```
IN [7]: NOTEBOOK.IPYNB

ampl.solve(solver=" highs")      # open source
ampl.solve(solver="gurobi")     # commercial
ampl.solve(solver="cplex")      # commercial
```

AMPL's MP driver also gives better behavior on logical and conditional constraints — your students' first instinct usually compiles.

SECTION 04

What AMPL ships in the toolkit

Tools in your students' hands

VS Code + AMPL extension

Syntax highlighting, debugging, integrated terminal, Git — the editor your students already know.

Jupyter Notebook + Colab

Narrative + code + output in one artifact. Zero-install via Colab kills the 'doesn't work on my machine' problem.

MO-Book + amplbot

Free, Colab-runnable optimization textbook + AI assistant trained on AMPL syntax.

LIVE DEMO

MO-Book Ch.1 — Production planning

<https://ampl.com/mo-book/notebooks/01/production-planning-advanced.html>



SECTION 04 • TOOLKIT

What just happened

- ✓ Same AMPL syntax — var, maximize, subj to
- ✓ Model is solver-agnostic
- ✓ Zero local install — your students did it from a browser

Default community license got us here. Let's talk about what's next.

SECTION 05

What this means for your team

AMPL travels with your students

<https://ampl.com/academia/>

A4A Academics

- Anyone with academic email
- 10 commercial + open source solvers
- No time limit, 12 months renewal
- Self-serve in Portal
- Free, individual use

A4C Courses

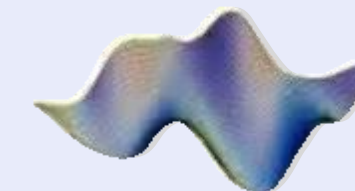
- Instructor-requested, distributable
- All commercial solvers, no fees
- Up to 8 months
- Shareable activation link
- Covers the whole class

<https://portal.ampl.com/user/ampl/request/ampl-for-courses/new>

ACS

- Recent grads (≤ 24 months)
- All commercial solvers
- 12 months · commercial OK
- Bridges classroom → industry

<https://portal.ampl.com/user/ampl/request/ampl-career-starter/new>



SECTION 05 · LICENSING

Where to start

- A** Portal home — sign in or create an account with your academic email
- B** A4A self-serve form — instant individual academic license
- C** A4C course request — instructor-only; returns a shareable activation link

portal.ampl.com/account/ampl/

COMING UP · PART 02

Part 2 — Hands-on

<https://github.com/ampl/modern-ampl>

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PRE-WORK CHECKLIST

- Laptop with Python 3.10+ (preferably)
- Visual Studio Code installed (recommended)
- A .mod file from your course
- Academic license already activated

WHAT YOU'LL DO

- Install amplpy + VS Code extension
- Run diet.mod as a baseline
- Switch HiGHS → Gurobi via your A4A